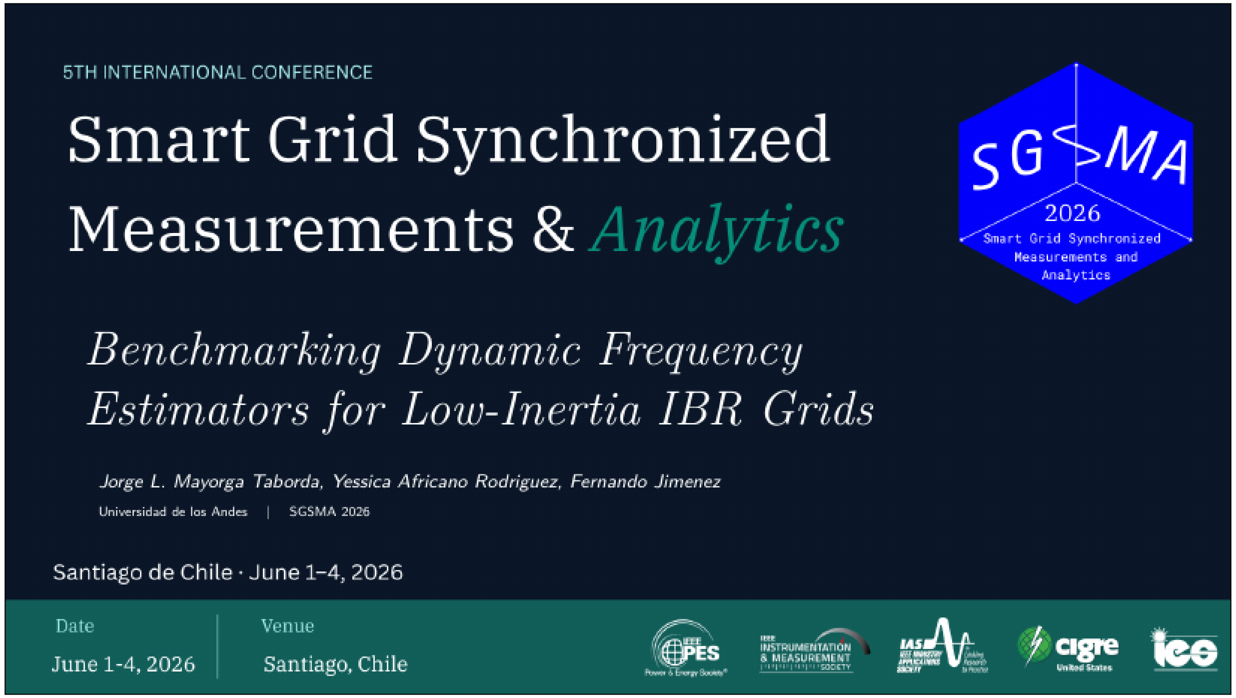


Title

Say:

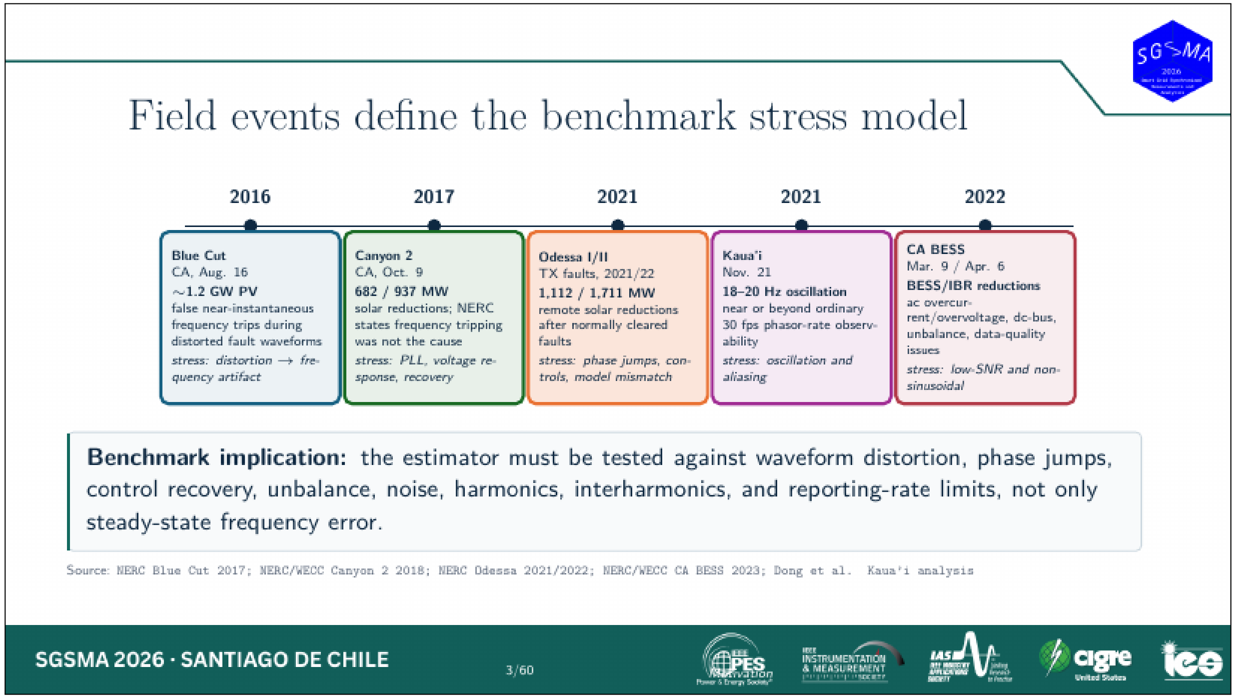
- Good morning, everyone. I am Jorge Mayorga, from Universidad de los Andes.
- I want to start with a deceptively simple question: in a grid run by inverters, which frequency estimator should we actually trust?
- The honest answer is, it depends. This talk makes 'it depends' precise.

→ *Let me start with why this is a problem worth solving.*

Key line: There is no universal best estimator; the right choice depends on what you measure.

Motivation (section)

Key line: Why this matters.



Field events define the stress model

Say:

- Real inverter disturbances are nothing like a clean sine wave.
- In events like Blue Cut, Odessa, and the Kauai oscillation, phase jumps, fast RoCoF, harmonics and noise all arrive together.
- So we let these real events define the stress cases our benchmark has to cover.

→ *And the tool we use to see these events also shapes what we conclude.*

Key line: Field events, not textbook signals, set the bar.

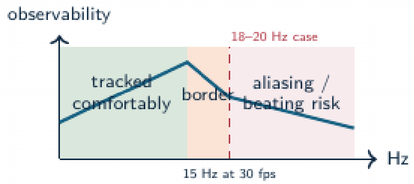
Standard PMU streams show consequences, not always causes

Fast fault and ride-through

IBR controls can react within cycles. A low-rate or heavily filtered frequency estimate may miss the subcycle cause while still showing the delayed consequence.

Oscillation near reporting-rate limits

The Kaua'i Island event on Nov. 21, 2021 exhibited 18–20 Hz oscillations after an oil-power-plant trip. A 30 frame/s phasor stream has a 15 Hz Nyquist limit for phasor-time variations, so oscillations near this range can alias or appear as beating.



Harmonics and interharmonics

Standard synchrophasor outputs summarize the fundamental component; harmonic and interharmonic contamination can still dominate estimator failure modes.

PMU streams show consequences

Say:

- A standard PMU gives us frequency and RoCoF.
- But during an inverter event, those numbers show the consequence, not always the cause, and the estimator inside the PMU decides what we see.

→ *So before we can analyze events, we have to ask which estimator to trust.*

Key line: Choosing the estimator is itself a measurement decision.

Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

Research problem

Say:

- And here is the gap: there is no agreed way to compare these estimators under inverter-specific stress.
- Compliance tests use clean signals, so they never tell us what happens on a real composite event.

→ *Our answer to that gap is one clear claim.*

Key line: We need a fair, inverter-relevant way to compare estimators.

Operating claim

- Frequency estimation under grid disturbances is a **multi-objective measurement problem**.
- Rankings change with the objective: frequency RMSE, RoCoF error, latency, CPU cost, and disturbance sensitivity produce different orders.
- IBR-era disturbances make this operational: distorted waveforms and fast controls can turn a measurement artifact into a ride-through or forensic diagnosis problem.
- Estimator choice is conditional: event family, objective metric, and compute budget determine the preferred method.

Measurement rule
Report error, transient exposure, settling, latency, and cost as separate outputs for each estimator.

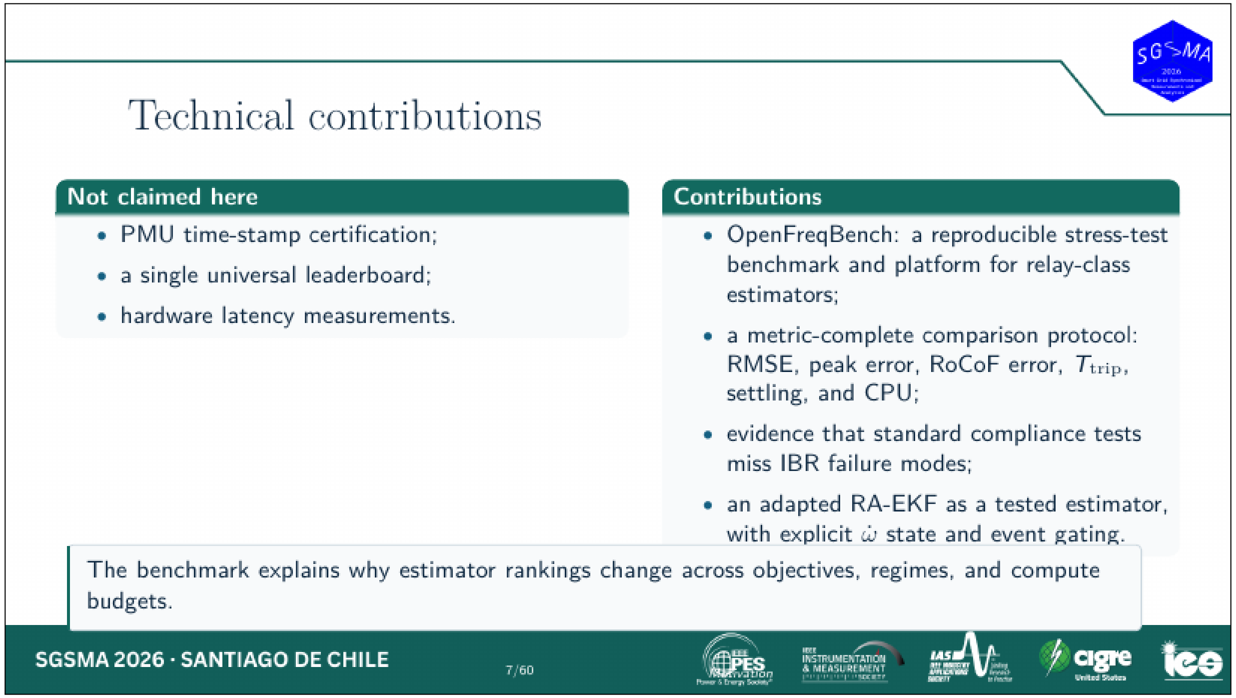
Operating claim

Say:

- Our claim is that estimator selection has to be conditional.
- It depends on the disturbance type, the metric you care about, and your compute budget.

→ *To support that claim we built three things.*

Key line: There is no single best estimator, only a best estimator for a given job.



Technical contributions

Say:

- First, an open benchmark platform with known-truth scenarios.
- Second, a common execution contract so all eighteen estimators are tested identically.
- And third, a metric vector that keeps accuracy, trip-risk, latency and cost separate.

→ *Let me give you a sense of the scale.*

Key line: Open platform, common contract, and multi-metric evidence.

Experiment scale

18
estimators

5
scenarios

4
families

grid
search

Central finding

No single estimator wins across objectives: the ranking changes with scenario, metric, and compute budget. Protection-class frequency measurement needs tests beyond compliance-style cases.

Dataset: 18 estimators (incl. legacy baselines) across 33 scenarios, $n = 5$ Monte Carlo per scenario, with standard and IBR-specific metrics, plus ATLAS severity sweeps.

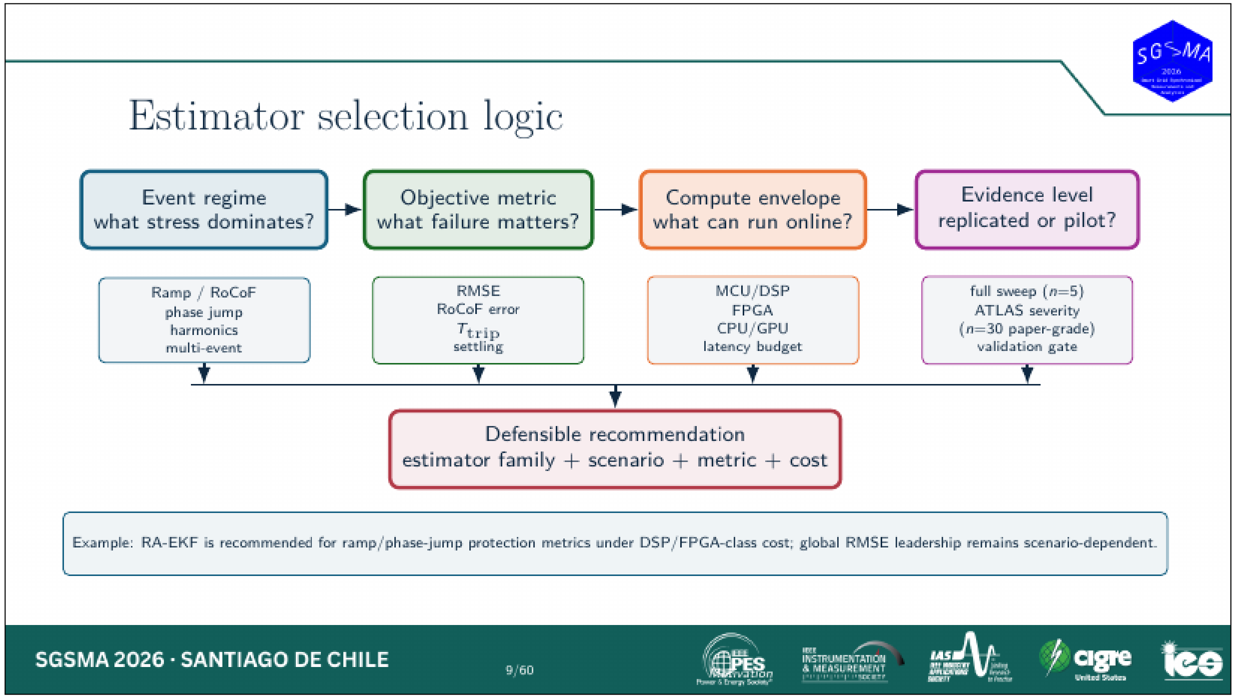
Experiment scale

Say:

- Eighteen estimators, thirty-three scenarios, five families of stress, with Monte-Carlo repetitions.
- On top of that, a severity-sweep engine we call ATLAS, and everything is reproducible from artifacts.

→ *And the logic for choosing between all these methods fits in one picture.*

Key line: Large, factorial, and fully reproducible.



Estimator selection logic (build-up)

Say:

- Selection comes down to three questions: which event, which metric, and which compute budget.
- Watch as each one is revealed, because each can point to a different method.

→ *With that logic in mind, let me show you the platform itself.*

Key line: This one diagram is the whole talk in miniature.

Benchmark and platform (section)

Key line: First, the benchmark and the platform.

Benchmark literature gap

Standards
PMU compliance, metrology, calibration

Field data
real events, PMU/DFR streams

Open platforms
OpenPMU, openPDC, event libraries

Single-method studies
detailed but narrow comparisons

Missing layer
many estimators, same truth, same metrics, same policy

Why existing evidence is not enough

Standards define requirements; field events define realism; open PMU ecosystems provide infrastructure. None alone answers which estimator family is defensible under the same disturbance, tuning policy, metric vector, and compute budget.

known truth → IBR stress → trip risk → CPU / latency

Estimator-level benchmark: one artifact contract for algorithms, scenarios, seeds, metrics, and plots

Result: controlled synthetic truth isolates estimator behavior under reproducible stress while field events define realism.

Benchmark literature gap

Say:

- Prior work usually compares a handful of estimators on one or two cases.
- Nobody has covered the full estimator space against composite inverter stress with reproducible metrics.

→ *So we built exactly that.*

Key line: That blank space on the map is what we fill.

Benchmark comparison axes

Evidence source	Known truth	IBR composite stress	Trip-risk	CPU/latency	Reproducible artifacts
IEC/IEEE-style compliance	Yes	Limited	No	No	Device-dependent
Field PMU/DFR data	No	Yes	Possible	No	Dataset-dependent
Single-method studies	Often	Partial	Partial	Partial	Often limited
Estimator benchmark	Yes	Yes	Yes	Yes	Yes

Measured outputs

The same voltage trace is scored by RMSE, peak error, RoCoF error, trip exposure, settling, CPU, and structural latency.

Interpretive effect

An estimator can pass standard cases and still fail a composite IBR event; it can also be accurate but too delayed or costly for protection-adjacent use.

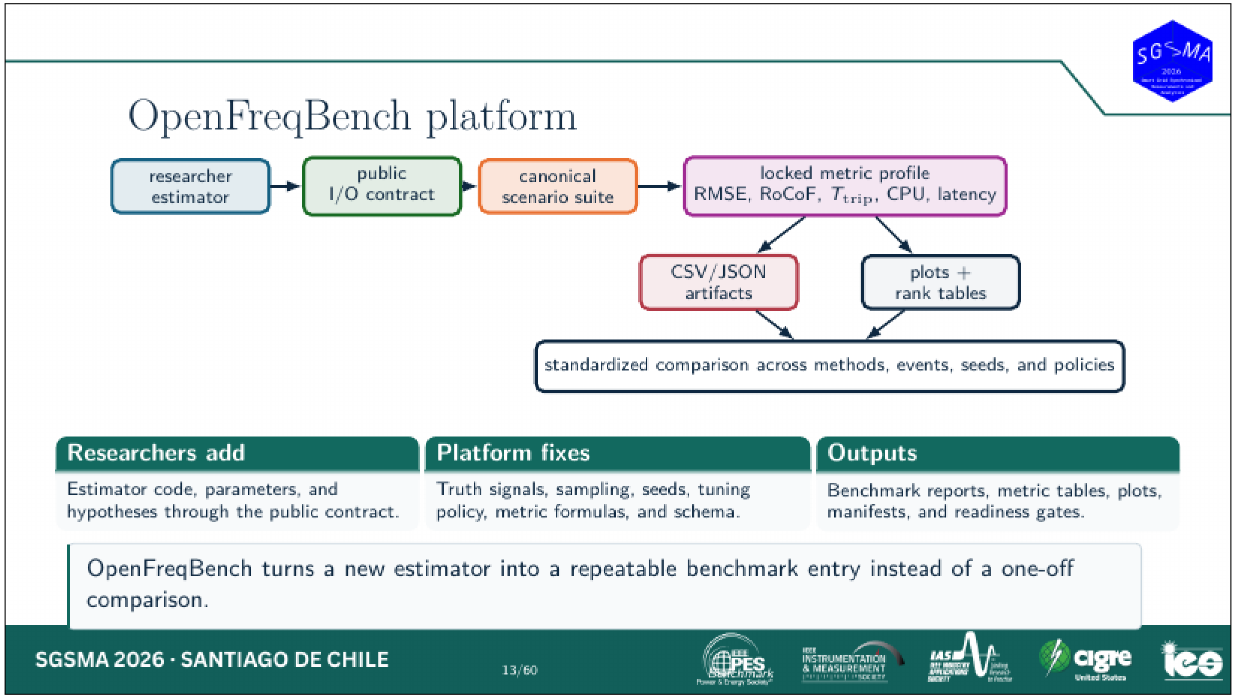
Benchmark comparison axes

Say:

- We compare on four axes at once: accuracy, robustness, computational cost, and protection trip exposure.
- The key point is that a method can win one axis and lose another.

→ *Here is the platform that measures all four.*

Key line: One number hides the trade-off; four axes reveal it.



OpenFreqBench platform

Say:

- OpenFreqBench has four parts: a scenario factory with known truth, a uniform estimator interface, a locked metric engine, and traceable artifacts.

→ *Let me show you how we keep the comparison fair.*

Key line: Known truth in, locked metrics out, everything traced.

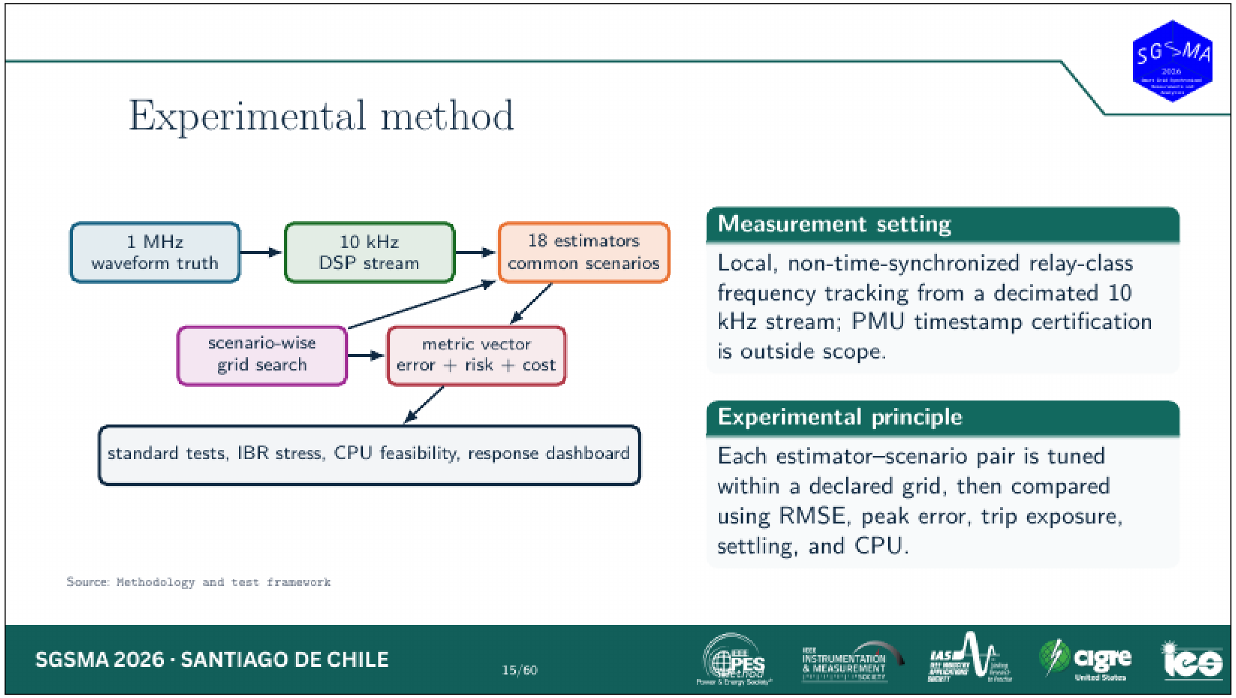
Experimental method (1)

Say:

- Fairness is the whole point: every estimator sees the exact same input trace, the same random seeds, and the same warm-up handling.

→ *And we are equally strict about tuning and timing.*

Key line: Fairness is enforced by construction, not by trust.



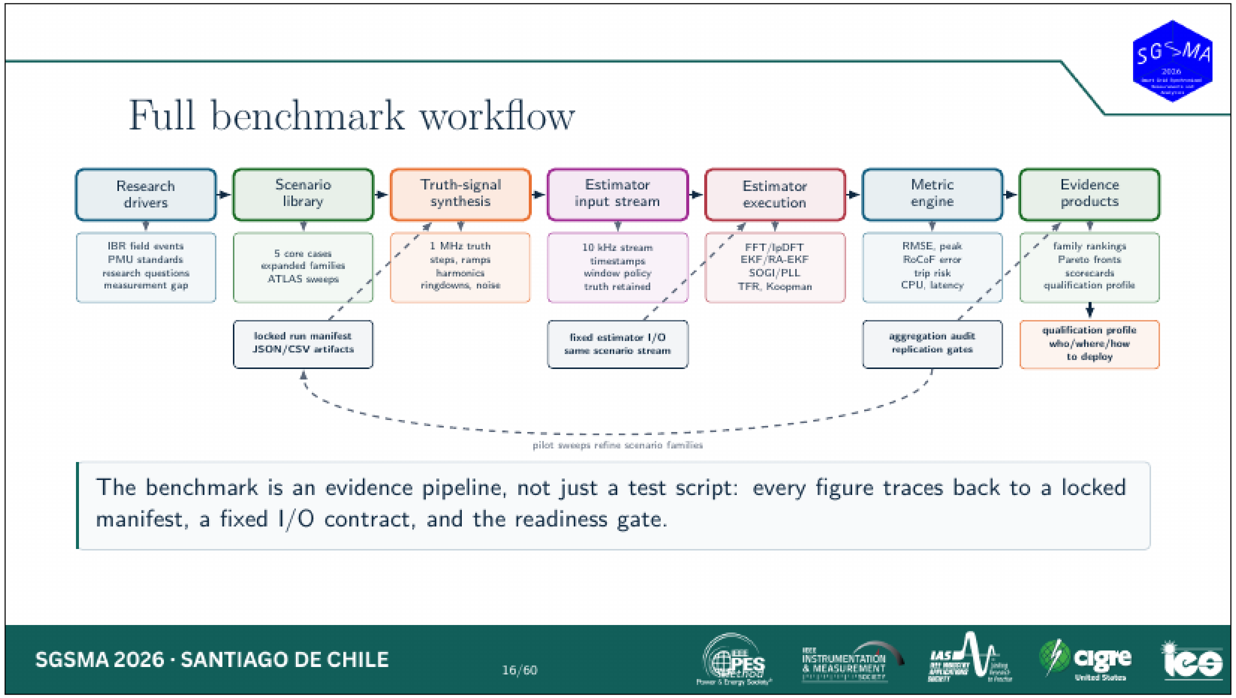
Experimental method (2)

Say:

- We tune under a fixed policy, we drop the warm-up region, and we measure CPU as a real output.

→ *Putting it together, this is the full pipeline.*

Key line: Nothing is hand-picked to flatter one method.



Full benchmark workflow

Say:

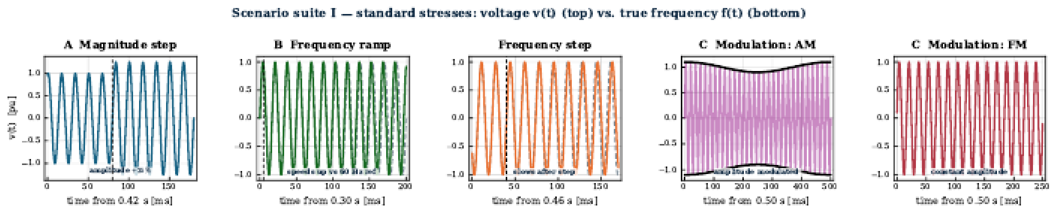
- Scenario, Monte-Carlo, estimator, locked metrics, then artifacts.
- Every single figure in this talk traces back to those artifacts.

→ *Let me show you the scenarios themselves.*

Key line: One reproducible pipeline, end to end.

Standard stresses: the waveform is controlled

CORE SCENARIO SUITE



Each standard test changes one physical ingredient at a time: magnitude, ramp, frequency step, AM, or FM.

Standard stresses: controlled waveform

Say:

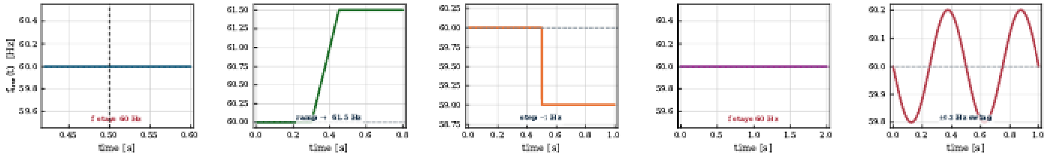
- We start with the standard stresses: a step, a ramp, and modulation.
- Because the waveform is fully controlled, we know the true frequency at every single sample.

→ *Here is what those look like.*

Key line: Known truth is what lets us score the error exactly.

Standard stresses: the frequency truth is simple

CORE SCENARIO SUITE



The clean reference traces make estimator failure visible: divergence is an algorithmic response, not ambiguity in ground truth.

Scenario suite I (dashboard)

Say:

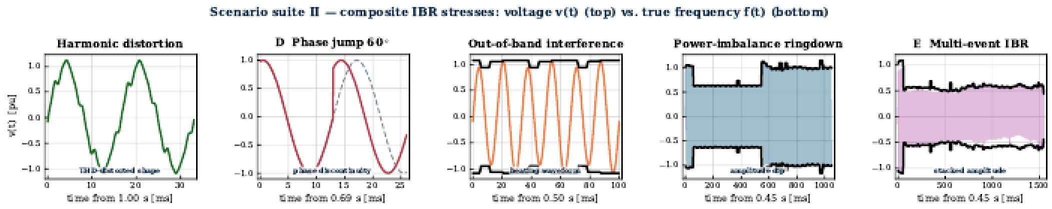
- Voltage on top, true frequency on the bottom, one column per scenario.
- Notice that for the magnitude step and the AM case, the frequency stays flat at sixty hertz, yet the estimators still react.

→ *Now we make it realistic.*

Key line: The stress is not always a frequency event, but estimators react anyway.

Composite IBR stresses: waveform physics are stacked

IBR STRESS SUITE



Composite cases combine harmonic distortion, phase jumps, out-of-band tones, ringdown recovery, and multi-event amplitude dynamics.

Composite IBR stresses

Say:

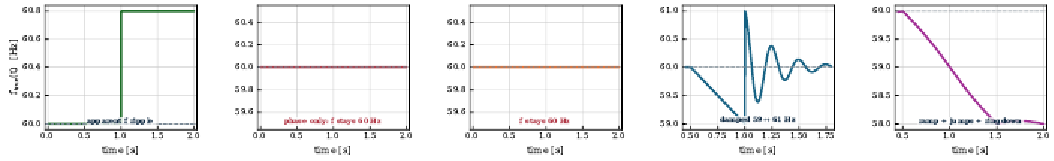
- Now we stack everything: harmonics, phase jumps, RoCoF segments and ringdown, all in one waveform.
- This is what a real inverter-dominated fault actually looks like.

→ *Here is that composite suite.*

Key line: Composite stress is the realistic, and the revealing, case.

Composite IBR stresses: frequency truth becomes diagnostic

IBR STRESS SUITE



The estimator must recover through nonstationary segments, not just report low steady-state error after the event.

Scenario suite II (dashboard)

Say:

- Same layout, voltage on top, frequency below, but these cases are deliberately harder than any compliance test.

→ *Before the results, a quick word on the methods and the metrics.*

Key line: If a method survives here, it has earned our trust.

Estimator families

Loop-based

PLL, SOGI-PLL, SOGI-FLL, Type-3 SOGI-PLL.

Window / spectral

IpDFT, TFT, ESPRIT, Prony, MUSIC.

Model-based / stochastic

EKF, UKF, RA-EKF, LKF, LKF2.

Adaptive and data-driven

RLS, Koopman (RK-DPMU).

Legacy baselines

TKEO and ZCD are retained as comparison references outside the recommended deployment set.

Interpretation rule

These families occupy different latency and stress-sensitivity positions. The comparison tracks where each family wins, degrades, and fails.

LKF and LKF2 follow H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

Estimator families

Say:

- The eighteen estimators fall into families: loop-based PLLs, window and spectral methods, model-based Kalman filters, adaptive, and data-driven.
- We analyze by family, because the family is what explains the behavior.

→ *And these are the numbers we score them on.*

Key line: Eighteen estimators, five families, one fair contract.

Metrics and decision variables

Error and recovery

$$e[k] = \widehat{f}[k] - f_{\text{true}}[k]$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_k e[k]^2}$$

$$E_{\max} = \max_k |e[k]|$$

Protection-oriented risk

$$T_{\text{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5 \text{ Hz}\}$$

$$t_s = \min\{t : \ |e[\tau]| \le \epsilon, \ \forall \tau \ge t\}$$

$$C_{\text{CPU}} = \text{mean time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

Metrics and decision variables

Say:

- We track RMSE for average error, peak error, and a trip-risk time, which counts how long the error stays above half a hertz.
- Plus settling time and CPU cost.

→ *Now let me show you what the data says.*

Key line: Half a hertz is our protection-oriented risk proxy.

Core evidence (section)

Key line: The core evidence: five scenarios, from standard to composite.

Standard scenario results

Method	Step	Ramp	Mod.
RA-EKF	0.000	**0.006**	0.037
SOGI-FLL	0.005	0.013	0.040
Koopman	0.003	0.017	0.044
TFT	0.001	0.019	0.046
UKF	0.000	0.017	0.055
LKF2	0.138	0.015	**0.028**
PLL	0.033	**1.47**	**0.227**
IpDFT	0.000	**14.4**	0.114
EKF	0.000	**37.1**	0.064

RMSE [Hz], mean over 5 seeds. Green: below 0.05 Hz guide; red: failure/divergence regime.

What the heatmap shows

The standard step is easy for most methods. The ramp is the separator: EKF and IpDFT can look excellent at a point and still diverge in another seed.

Operational consequence

Robustness is not the same as point accuracy. A standard scenario can pass while the same estimator remains unsafe for ramp-like IBR dynamics.

Result: RA-EKF, SOGI-FLL, Koopman, and TFT remain below 0.05 Hz across the three standard cases; EKF and IpDFT fail through intermittent ramp divergence.

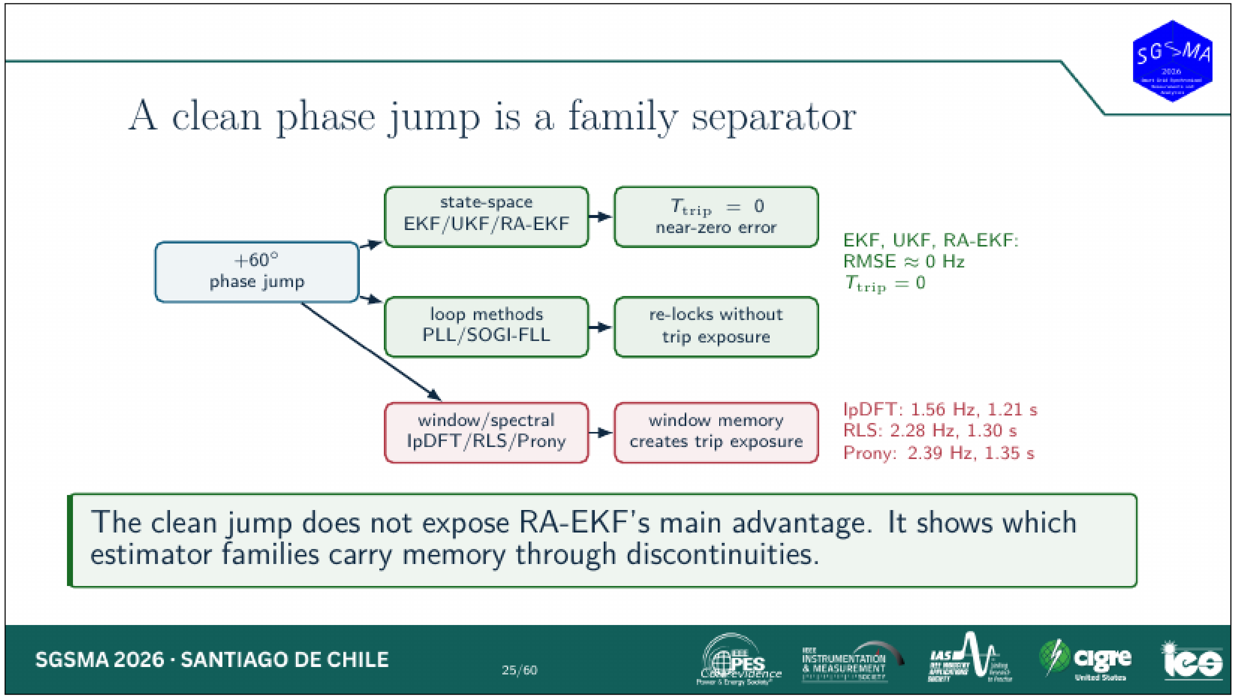
Standard scenario results (build-up)

Say:

- On the standard cases, most methods are excellent, all below fifty millihertz.
- CLICK. But watch the ramp: the textbook EKF and IpDFT intermittently blow up, to thirty-seven and fourteen hertz.

→ *So a phase jump should be even worse, right? Let me show you.*

Key line: Robustness, not point accuracy, is what separates the families.



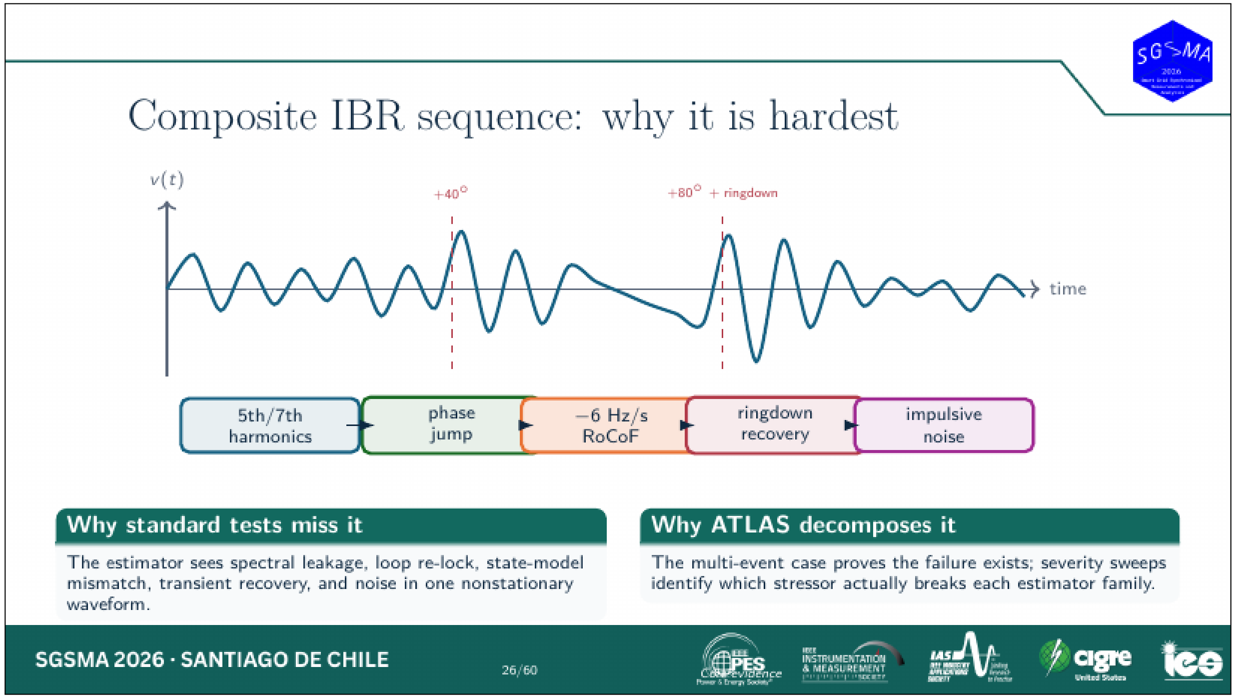
Scenario D: phase jump separates families

Say:

- A clean sixty-degree phase jump cleanly splits the families.
- The state-space filters and the PLLs recover with zero trip exposure.
- The windowed and spectral methods carry over a second of exposure.

→ *But a single jump is not the real challenge.*

Key line: Surprisingly, the phase jump does not break the EKF.



Composite IBR sequence: hardest case

Say:

- Scenario E stacks it all: harmonics, two phase jumps, a RoCoF segment, ringdown, and impulsive noise.
- Spectral leakage, loop re-lock and model mismatch all hit at the same time.

→ *And here the leaderboard falls apart.*

Key line: This is the hardest and most realistic case in the suite.

Scenario E: the leaderboard breaks

Method	RMSE [Hz]	T_{trip} [s]	CPU [μs]
UKF	**1.09**	**0.78**	14
EKF	1.13	0.87	11
SOGI-PLL	1.14	**0.78**	11
SOGI-FLL	1.14	0.84	**9**
IpDFT	1.14	1.35	17
RA-EKF	1.16	0.92	15
Koopman	1.90	0.84	**15 241**
ZCD	**426**	1.55	7

No accurate winner

Under the stacked sequence every usable method collapses to $\approx$ 1.1 Hz RMSE; ZCD diverges catastrophically (426 Hz). High accuracy is simply not available here.

Scalar ranking failure

UKF has the lowest RMSE and trip exposure, SOGI-FLL the lowest usable CPU, Koopman is 1000× costlier; RA-EKF sits mid-pack but never diverges.

Source: full_mc_benchmark, n = 5: IBR_Multi_Event

Scenario E: the leaderboard breaks

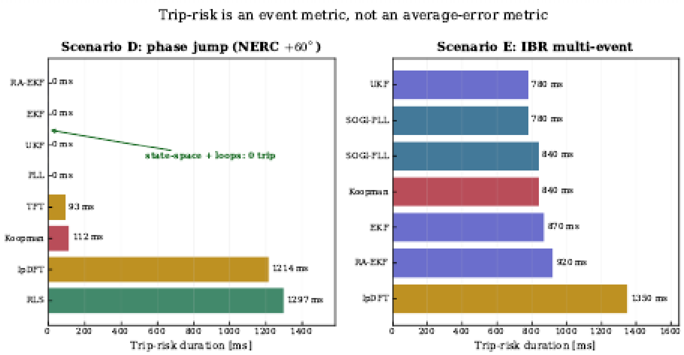
Say:

- Every usable method collapses to about one-point-one hertz; ZCD diverges to four hundred hertz.
- And the rankings by RMSE, by trip-risk, and by CPU all disagree with each other.

→ *Let me make the trip-risk part visual.*

Key line: Under stacked stress there is simply no accurate winner.

Trip-risk evidence: core stress cases



Result reading

- Scenario D: state-space filters and PLL keep $T_{\text{trip}} = 0$; windowed/spectral methods (IpDFT, RLS) exceed 1.2 s.
- Scenario E: *every* method carries 0.8–1.4 s exposure — composite stress is unavoidable.
- The trip-exposure ranking is not the RMSE ranking.

Result: T_{trip} separates protection exposure from average error.

Trip-risk evidence (figure)

Say:

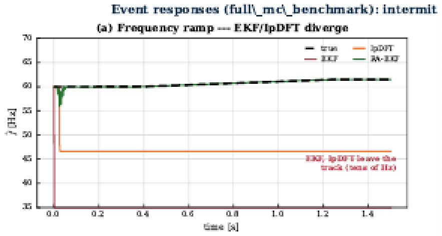
- In Scenario D the state-space filters hold zero exposure while windowed methods pass 1.2 seconds.
- In Scenario E, everyone carries exposure.

→ *And you can watch the divergence happen in real time.*

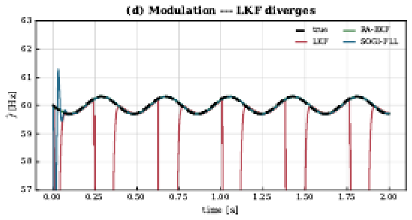
Key line: Trip-risk is a different ranking from RMSE.

The failures are visible in the trace, not only in the table

RAMP DIVERGENCE



MODULATION DIVERGENCE



Failure signature
EKF and IpDFT leave the true ramp almost immediately; after the event, the estimate is no longer operationally usable.

Failure signature
LKF stays plausible between collapses, then repeatedly drops out of the usable frequency band.

Event responses (figure)

Say:

- True frequency in black, and each estimator trying to track it.
- You can literally see EKF and IpDFT run away on the ramp, while RA-EKF stays locked.

→ *So what does all of this mean if you have to choose?*

Key line: This is the intermittent divergence, seen with your own eyes.

Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	State-space + SOGI-FLL near-zero; LKF2 best on mod	Step ≈ 0; mod 0.028–0.044 Hz	Clean cases do not predict ramp/event survival
Ramp and RoCoF tracking	RA-EKF is the most robust ramp tracker	0.006 Hz vs EKF 37 Hz, IpDFT 14 Hz (diverge)	Divergence is intermittent (seed-dependent)
Phase jump (Scenario D)	State-space filters recover with zero trip	EKF/UKF/RA-EKF $T_{\mathrm{trip}} = 0$; IpDFT 1.21 s	Windowed/spectral carry the exposure
Multi-event RMSE	No accurate method — all ≈ 1.1 Hz	UKF 1.09 Hz best; ZCD 426 Hz diverges	RMSE hides trip-risk and 1000× CPU spread
Embedded feasibility	PLL, SOGI-FLL, EKF, RA-EKF remain plausible	9–17 μs/sample; Koopman ∼15,000	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

Result comparison: winners by question

Say:

- Ask for clean accuracy and you get the state-space and SOGI-FLL.
- Ask for ramp robustness and you get RA-EKF. Ask about multi-event and there is no winner. Ask about feasibility and the expensive methods drop out.

→ *Feasibility brings us to cost.*

Key line: Different question, different winner, every time.

Computational feasibility

Method	Time/sample	Family	Target
ZCD	7 μs	Loop	MCU
SOGI-FLL	9 μs	Loop	MCU
PLL	9 μs	Loop	MCU
EKF	11 μs	Model	MCU/DSP
UKF	13 μs	Model	DSP
RA-EKF	**15 μs**	**Model**	**DSP**
IpDFT	17 μs	Window	DSP/FPGA
ESPRIT	$\approx$1,400 μs	Window	CPU
Koopman	$\approx$15,000 μs	Data-driven	CPU/GPU

Source: `full_mc_benchmark`, $n = 5$: mean `m13_cpu_time_us`

Deployment interpretation

The Python baseline separates feasible estimator families from clearly prohibitive ones; hardware latency requires platform timing.

Key comparison

The state-space and loop families all run at ~10–17 μs/sample (embedded-feasible); spectral (ESPRIT, MUSIC) and data-driven (Koopman) methods are 100–1000× costlier and data-dependent.

Computational feasibility

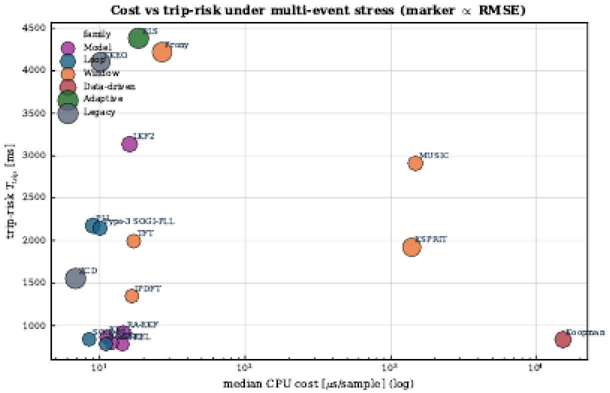
Say:

- The loop and state-space methods run in nine to seventeen microseconds per sample, which is embedded-feasible.
- The spectral and data-driven methods are a hundred to a thousand times slower.

→ *Put cost and risk together and the picture is clear.*

Key line: Cost alone already removes several methods from a relay.

Deployment limits



Deployment interpretation

- Deployment combines RMSE, trip exposure, CPU, and divergence robustness.
- State-space + loop methods cluster at low cost ($\leq 17\,\mu$s) and low trip.
- Spectral/data-driven methods trade lower error for 100–1000× higher CPU.
- RA-EKF stays low-cost with no ramp divergence in this run.

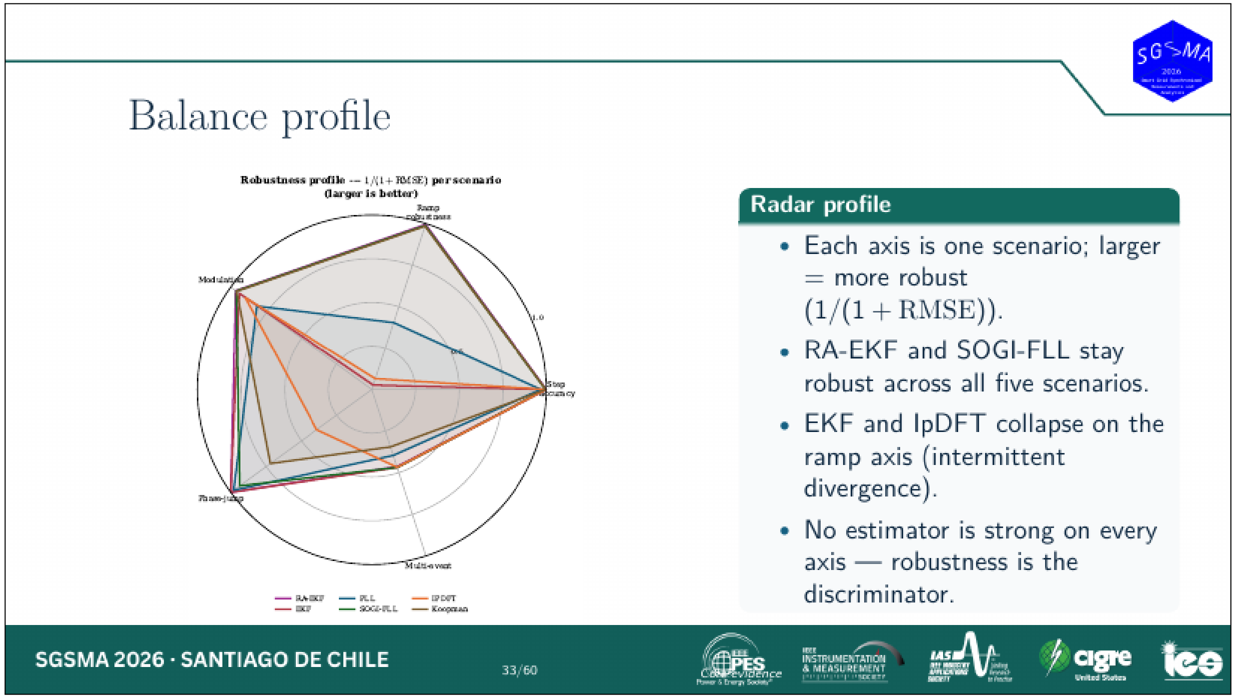
Deployment limits (figure)

Say:

- Deployment is the combination of accuracy, trip exposure, cost, and divergence-robustness.
- And the low-cost cluster, the state-space and loop methods, is also the low-trip cluster.

→ *We can summarize each method's balance in one shape.*

Key line: The practical winners sit together at low cost and low trip.



Balance profile (radar)

Say:

- Each axis is one scenario, and bigger means more robust.
- RA-EKF and SOGI-FLL stay robust on every axis; EKF and IpDFT collapse on the ramp axis.

→ *Now, the most important slide about compliance testing.*

Key line: No method is strong on every axis.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Jump	Multi
Koopman	P	P	P	F	F
TFT	P	P	P	F*	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	P	P	P	F
PLL	P	F	F*	P	F
EKF	P	F	P	P	F
UKF	P	P	P	P	F
RA-EKF	P	P	P	P	F
LKF2	F*	P	P	P	F

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/T_{trip}/settling threshold | **F*** borderline.

Result: Step, ramp, and modulation compliance does not predict phase-jump or multi-event readiness.

Compliance heatmap (build-up)

Say:

- Look at the standard columns first: almost everyone passes step, ramp, and modulation.
- CLICK. But the same methods fail on phase jump and multi-event.

→ *Let me distill that into our empirical claims.*

Key line: Passing the standard tests does not predict event readiness.

Empirical claims

Protection behavior

Divergence, not phase jumps, is the dominant failure mode. On the ramp, EKF and IpDFT intermittently diverge (37 and 14 Hz mean); RA-EKF stays at 0.006 Hz. A clean phase jump is recovered by every state-space filter with $T_{\text{trip}} = 0$.

Metric disagreement

RMSE, trip-risk, and CPU select different winners. Under multi-event stress UKF has the lowest RMSE/trip, SOGI-FLL the lowest usable CPU, and Koopman is 1000× costlier.

Standard-test limitation

Simple-case pass/fail leaves event readiness unresolved. The heatmap shows methods that pass Step/Ramp/Mod but fail islanding or multi-event IBR stress.

Deployability

Feasibility changes the accuracy ranking. RA-EKF runs at ∼15 μs/sample; the spectral and data-driven methods are 100–1000× costlier and sit outside relay-class timing.

Guardrail: report scenario, metric, and compute envelope before naming a preferred estimator.

Empirical claims

Say:

- Four claims. Divergence, not phase jumps, is the dominant failure mode.
- Pass-fail leaves event readiness unresolved; the three metrics pick different winners; and feasibility changes the ranking.

→ *Now, does this pattern survive when we scale up?*

Key line: Four claims, one theme: always report the conditions.

Expanded and ATLAS (section)

Key line: Does the pattern hold across families and severity? That is ATLAS.

Expanded stress analysis

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Monte-Carlo benchmark (full_mc)

- 18 estimators across 5 families.
- 33 scenarios (A–E plus severity variants).
- $n = 5$ Monte Carlo seeds per scenario.
- Standard, phase-jump, multi-event, and CPU metrics.

ATLAS severity sweeps

- Severity, sign, and policy curves per stress family.
- Fixed-policy, paper-grade $n = 30$ across all 9 required sweeps (both signs).
- Harmonic, RoCoF, phase-jump, and noise atlases.
- Tests sensitivity, scaling, and family stability.

Two evidence layers on the same truth: a replicated Monte-Carlo benchmark, plus ATLAS severity sweeps that trace where each estimator family degrades.

Expanded stress analysis

Say:

- We scale up to a full Monte-Carlo benchmark: eighteen estimators, thirty-three scenarios, five seeds.
- And then the ATLAS severity sweeps on top.

→ *First, do the family winners stay fixed?*

Key line: We scale the test, we do not just repeat it.

Expanded benchmark: family winners

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Family	Count	RMSE leader	RMSE	Trip-risk leader
IEEE ramps	9	RA-EKF	11 mHz	0 s (state-space)
Magnitude steps	7	UKF	<1 mHz	0 s (state-space)
Modulation	3	LKF2	15 mHz	0 s (state-space)
Phase jumps	3	UKF	<1 mHz	0 s (state-space)
Frequency step	1	RA-EKF	52 mHz	RA-EKF, 3 ms
OOB interference	1	UKF	<1 mHz	0 s (state-space)
IBR harmonics	3	SOGI-FLL	61 mHz	SOGI-PLL, 10 ms
IBR ringdown	4	LKF2	202 mHz	ESPRIT, 87 ms
IBR multi-event	1	UKF	1.09 Hz	UKF, 780 ms

The full Monte-Carlo sweep confirms the central pattern: the RMSE leader rotates (RA-EKF, UKF, LKF2, SOGI-FLL) by disturbance family, and error grows from sub-mHz on clean cases to ~1 Hz under multi-event stress.

Expanded benchmark: family winners

Say:

- They do not. The winner rotates: one method leads the step, another the ramp, another harmonics, another the multi-event.

→ *ATLAS lets us see why, by sweeping severity.*

Key line: The leaderboard is regime-dependent, never absolute.

ATLAS severity analysis

PAPER-GRADE SWEEP: $n = 30$, **fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)**

Core cases → ATLAS severity curves → sign asymmetry → fixed vs oracle policy → validation gate

- Core cases: A–E failure mode
- ATLAS severity curves: When does error start to grow?
- sign asymmetry: Are positive and negative events symmetric?
- fixed vs oracle policy: Does tuning policy change the rank?
- validation gate: Which results replicate?

ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Source: docs/ATLAS_PIPELINE.md; docs/ATLAS_METHOD_AUDIT.md

ATLAS severity analysis

Say:

- ATLAS sweeps each stress by severity and by sign, under a fixed policy, at thirty Monte-Carlo runs.
- This is now paper-grade across all nine required sweeps.

→ *And the headline result is striking.*

Key line: This is paper-grade severity evidence, not a pilot.

ATLAS: metric choice changes the answer

Say:

- Across a hundred and thirteen severity levels, the preferred estimator changes when the objective changes.
- In eighty-three of them, RMSE and trip-risk point to different methods.

→ *ATLAS also tells us where each method gives up.*

Key line: The output is a selection map, not a leaderboard.

ATLAS result: metric choice changes the answer

PAPER-GRADE SWEEP: $n = 30$, **fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)**

Comparison	Different winner	Levels tested
RMSE vs trip-risk	83	113
RMSE vs RoCoF max error	55	113
RMSE vs RoCoF RMS error	50	113

Winner concentration

RMSE is led mostly by EKF and ESPRIT. Trip-risk is led mostly by Koopman, EKF, SOGI-FLL, ZCD, and ESPRIT. CPU is always led by ZCD.

Why this matters

For protection-adjacent use, a low average error is not enough. The estimator must avoid long deadband exposure, track RoCoF, and remain computable.

Operational implication

The benchmark output is not a leaderboard. It is a metric-conditioned selection map.

Result: across 113 severity levels, the preferred estimator changes when the objective changes.

Source: `atlas-papergrade-missing-v1/global_metrics_report.csv`

ATLAS result: where estimators stop being valid

PAPER-GRADE SWEEP: $n = 30$, **fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)**

Stress family	Maximum tested level	Methods below 0.05 Hz	Interpretation
Magnitude step	$\pm 90\%$ sag/swell	2	EKF and RA-EKF survive both signs.
AM modulation	10 Hz envelope	10	AM alone is not the hardest modulation case.
FM modulation	10 Hz modulation	0	FM is qualitatively harder than AM.
RoCoF ramp	± 50 Hz/s	0	Severe RoCoF requires a separate qualification class.
Phase jump	$\pm 60^\circ$	0	Single jump severity already breaks the 0.05 Hz guide.
Harmonics	30% THD	0	Low-THD winners do not survive extreme THD.
Interharmonics	20% at 75 Hz	0	Integer-harmonic tests are not sufficient.
Broadband noise	$\sigma = 0.1$ pu	0	Noise robustness is a first-class stress dimension.

ATLAS: where estimators stop being valid

Say:

- For each method, ATLAS finds the severity at which it crosses into failure.

→ *Let me show you a few of the most surprising sweeps.*

Key line: That validity boundary is different for every family.

AM, FM, and interharmonics split the story

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Stress	Winner pattern	Threshold behavior
AM modulation	EKF wins slow, reference, fast, and severe regions.	Ten methods stay below 0.05 Hz through 10 Hz envelope modulation.
FM modulation	TFT leads slow; ESPRIT leads reference/fast/severe; SOGI-PLL reaches the best threshold.	No method stays below 0.05 Hz at 10 Hz FM modulation.
Interharmonics	EKF wins trace, reference, high, and severe levels.	EKF crosses the 0.05 Hz guide around 8%; most other families cross earlier.

Implication

AM, FM, integer harmonics, and interharmonics should not be collapsed into one "distortion" label. They select different estimator families.

Decomposition insight

A composite IBR trace can contain AM, FM, harmonics, interharmonics, noise, and phase steps. ATLAS isolates which stressor changes the ranking.

AM, FM and interharmonics

Say:

- Modulation actually splits the story: AM is easy, and the model-based methods win.
- FM and interharmonics are harder, and the spectral methods take over.

→ *Now the four findings I most want you to remember.*

Key line: Even inside modulation, the winner switches.

When the textbook filter diverges

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Per-seed frequency RMSE [Hz] on the IEEE frequency-ramp scenario (5 Monte Carlo seeds):

Estimator	seed 1	seed 2	seed 3	seed 4	seed 5	trip-risk
EKF (textbook)	0.016	63.7	0.017	121.5	0.027	up to 1.35 s
RA-EKF	0.0055	0.0062	0.0083	0.0045	0.0058	0 s

What happens

The standard EKF tracks most seeds but **intermittently diverges** to tens of Hz. It is *not* a severity threshold: the worst seed (121 Hz) had a low ramp rate (1.7 Hz/s) while the steepest (5 Hz/s) tracked cleanly. The failure is triggered by onset/phase combinations — seed-dependent and silent.

Why RA-EKF matters here

Covariance scaling and event gating remove the divergence: stable on **every** seed, zero trip exposure. This — not a universal RMSE win — is its defensible value.

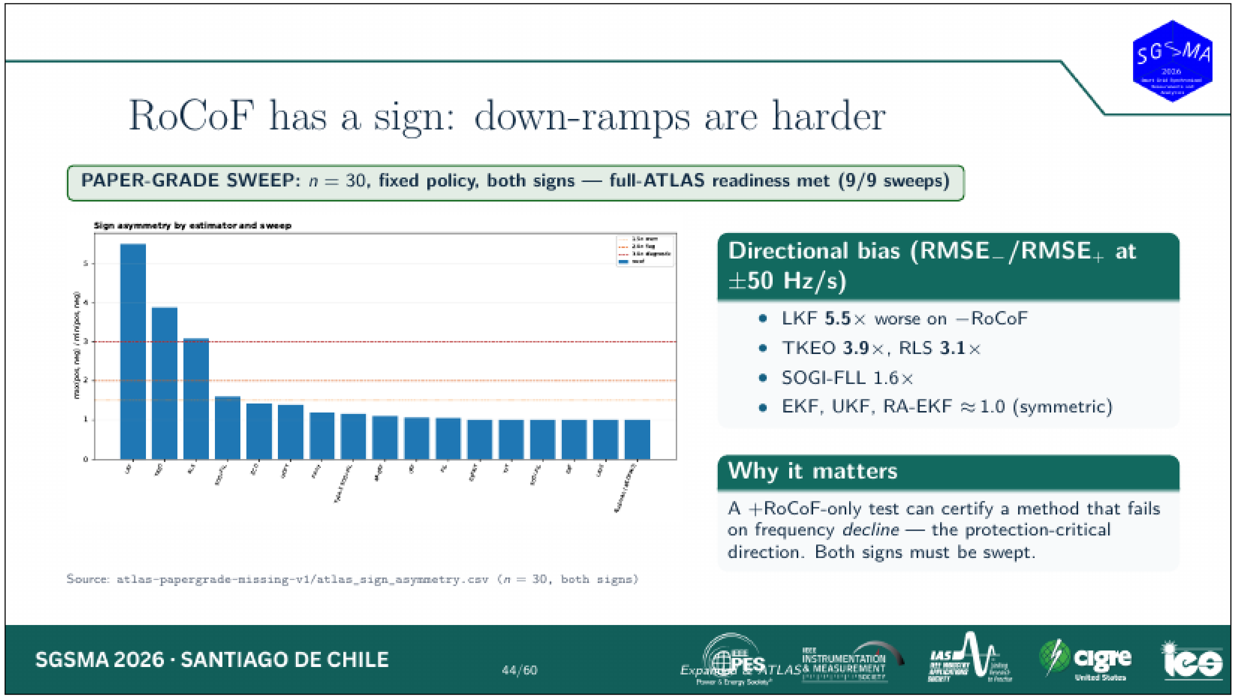
When the textbook filter diverges

Say:

- The textbook EKF tracks most seeds, then intermittently runs away to tens of hertz.
- And it is not severity-driven: the worst seed actually had the gentlest ramp.

→ *The next finding is about direction.*

Key line: The robustified RA-EKF removes the divergence on every seed.



RoCoF has a sign

Say:

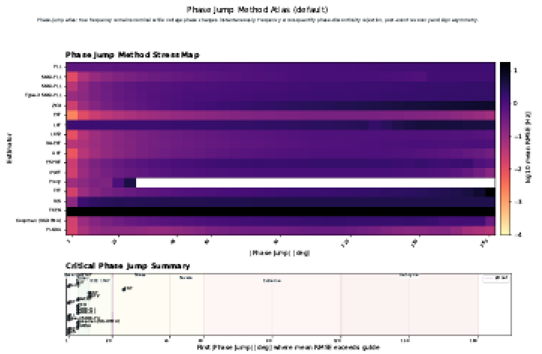
- RoCoF has a sign, and down-ramps are harder than up-ramps.
- LKF is five-and-a-half times worse on a falling frequency; TKEO almost four times worse.

→ *Severity can also surprise us in the other direction.*

Key line: A rising-only test can certify a method that fails when frequency falls, the protection-critical case.

Phase-jump severity is non-monotone

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Worst case is not 180°

Most methods peak near 90–120° and *recover* toward 180° (a half-cycle jump partly self-cancels). EKF peaks at only **0.25 Hz @ 115°**, back to 0.07 Hz @ 180°.

The TFT cliff

One exception: TFT rises gently, then jumps **2.5 → 21.9 Hz** between 150° and 180° — a phase-reversal singularity.

Result: Severity is non-monotone; sweep the jump angle rather than qualify one nominal point.

Phase-jump severity is non-monotone

Say:

- Phase-jump severity is non-monotone: the worst case is not a hundred and eighty degrees.
- Most methods peak around ninety to a hundred and twenty degrees and then recover, because a half-cycle jump partly cancels itself.

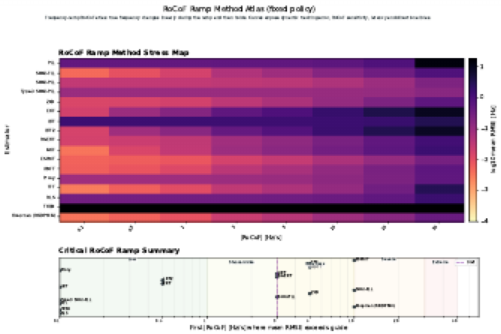
→ *Severity also reshuffles the winners completely.*

Key line: Bigger is not always worse, so you must sweep the range.

No estimator owns the whole stress space

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

RoCoF regime	Best	med. RMSE
Low (0.1–1 Hz/s)	Koopman	4.1 mHz
Standard (1–3)	ESPRIT	7.6 mHz
IBR stress (3–10)	ESPRIT	22.3 mHz
Severe (10–30)	SOGI-PLL	70.4 mHz



Across stress types

EKF wins phase jumps, the magnitude step and interharmonics; ESPRIT/Koopman win the RoCoF and frequency-step regimes; ZCD wins low-THD harmonics; SOGI-PLL leads severe RoCoF. No method leads everywhere.

The deliverable is a selection map: event class × metric × compute budget.

No universal winner (map)

Say:

- This map says everything: the champion rotates by RoCoF regime.
- Koopman at low rates, ESPRIT in the middle, SOGI-PLL when it gets severe.

→ *And one estimator manages to be both the best and the worst.*

Key line: No universal winner; the deliverable is the map itself.

ZCD is a narrow instrument, not a general estimator

PAPER-GRADE SWEEP: $n = 30$**, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)**

Low-stress champion

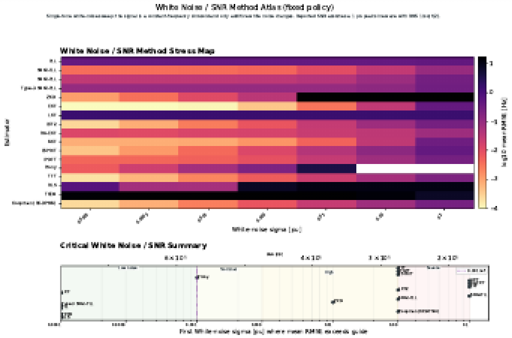
Up to ~15% THD, zero-crossing detection gives the *lowest* error of any method — **0.27 mHz** at 1% THD, 2.2 mHz at 15%; and just 1.1 mHz at low broadband noise.

Broadband-noise cliff

Same method: 34 mHz at $\sigma = 0.003$ pu $\rightarrow$ **3112 Hz** at $\sigma = 0.1$ pu. EKF stays ≤ 0.43 Hz across the same sweep.

Harmonic cliff

Same method again: 2.2 mHz at 15% THD $\rightarrow$ **39 Hz** at 20% $\rightarrow$ **86 Hz** at 30%.



The ZCD paradox (build-up)

Say:

- Below fifteen percent harmonic distortion, zero-crossing detection is the most accurate method we have.
- CLICK. Add broadband noise and it explodes to three thousand hertz.
- CLICK. Push past twenty percent distortion and it collapses to tens of hertz.

→ *The same asymmetry shows up in voltage events.*

Key line: The same estimator is the best and the worst, depending on the stress.

Voltage sags and swells are not symmetric

PAPER-GRADE SWEEP: $n = 30$, **fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)**

Estimator at ±90%	Sag to 0.1 pu	Swell to 1.9 pu
EKF RMSE	47 mHz	14 mHz
RA-EKF RMSE	49 mHz	30 mHz
ZCD RMSE	374 Hz	9.4 mHz
ZCD peak error	2856 Hz	32 mHz

Mechanism

Zero-cross timing error scales roughly as $\sigma_n/(A\omega)$. A deep sag lowers zero-cross slope and makes false crossings likely when absolute noise is fixed.

What to claim

ZCD is amplitude-invariant only in the noise-free ideal case. Under fixed absolute noise, deep voltage sags create a low-SNR crossing problem.

Benchmark consequence

Sag and swell signs must be swept separately; a positive-only magnitude-step test can hide the worst case.

Source: atlas-papergrade-missing-v1/global_metrics_report.csv; src/estimators/zcd.py

Sags and swells are not symmetric

Say:

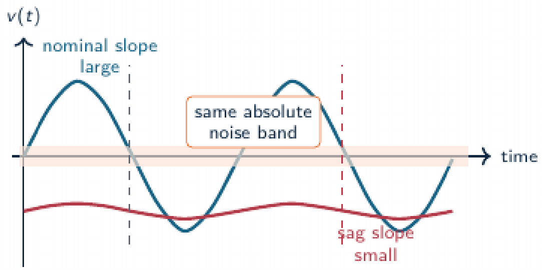
- Voltage magnitude is not symmetric either: a deep sag is far harder than a swell.
- ZCD goes from millihertz on a swell to hundreds of hertz on a deep sag.

→ *And there is a clean physical reason.*

Key line: Direction matters; you have to sweep both signs.

Why a voltage sag breaks zero-crossing detection

PAPER-GRADE SWEEP: $n = 30$, **fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)**



Timing sensitivity

$$\Delta t \approx \frac{n(t_0)}{\left.\frac{dv}{dt}\right|_{t_0}} \qquad \left.\frac{dv}{dt}\right|_0 \approx A\omega$$

When the voltage amplitude A collapses, the same noise produces much larger crossing-time error.

Estimator consequence

ZCD is elegant at clean high-SNR crossings, but deep sags convert amplitude loss into apparent frequency jitter and false crossings.

Why a sag breaks zero-crossing

Say:

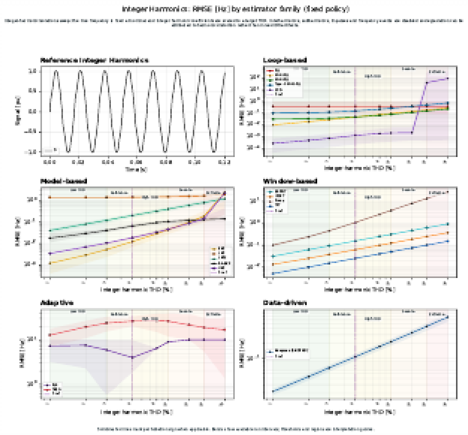
- The timing error of a zero-crossing scales with noise over amplitude times frequency.
- A deep sag flattens the slope right at the crossing, so noise creates false crossings.

→ *Harmonics tell a similar family-by-family story.*

Key line: This is physics, not a software bug.

Harmonic degradation differs by estimator family

PAPER-GRADE SWEEP: $n = 30$, **fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)**



Result

- Loop, window, model, and adaptive families degrade with THD at different rates.
- Best low-THD accuracy does not predict high-THD survival.
- Loop methods (ZCD) win at low THD but collapse above ∼15%; window methods (TFT) degrade gently and lead at extreme THD.

Harmonic degradation by family

Say:

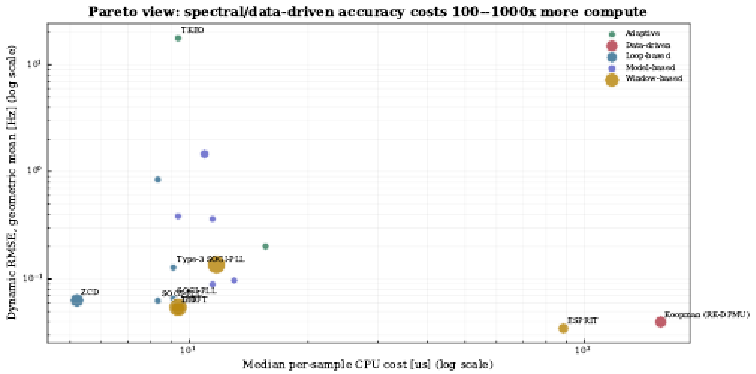
- Harmonic degradation depends on the family: loop methods like ZCD win at low distortion but collapse above fifteen percent.
- Window methods degrade gently and end up leading at extreme distortion.

→ *And cost is the last dimension.*

Key line: Low-distortion accuracy does not predict high-distortion survival.

Accuracy costs 100–1000× more compute

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario



Low-cost core
PLL, SOGI-FLL, EKF, and RA-EKF sit in the deployable compute range.

Accuracy premium
Spectral and data-driven methods can move right by 100–1000× CPU cost.

CPU-accuracy Pareto

Say:

- On the cost-accuracy Pareto, the spectral and data-driven methods buy a little accuracy for a hundred to a thousand times more compute.

→ *So, what do we actually deploy?*

Key line: For relay-class timing, that trade is usually not worth it.

Findings and deployment (section)

Key line: So what do we actually deploy?

Three findings

1. Metric-conditioned
The RMSE leader often differs from the trip-risk and RoCoF leader across ATLAS severity levels.

2. Stress-bounded
AM remains survivable for many methods; severe FM, RoCoF, interharmonics, noise, and extreme THD break the 0.05 Hz guide.

3. Robustness over point accuracy
EKF, UKF, and IpDFT can diverge on ramps; RA-EKF keeps the ramp family stable across seeds.

> Dynamic frequency estimators should be qualified by event family, sign, severity, recovery, trip exposure, and compute envelope.

Three findings

Say:

- Three findings. One, there is a latency-versus-stress trade-off.
- Two, robustness beats point accuracy: RA-EKF's explicit RoCoF state keeps it stable exactly where EKF and IpDFT diverge.
- Three, compliance tests are not enough for inverter events.

→ *Let me be honest about the limits.*

Key line: Robustness, not accuracy, is the deployment discriminator.

Validity limits and next steps

Validation limits

- Limited to the reported scenario set.
- CPU is a software-level cost indicator; hardware latency requires platform timing.
- The study is simulation-only.
- Scenario-wise tuning can affect rank stability.

Validation program

- Hardware-in-the-loop validation.
- Wider converter-control interaction coverage.
- Sensitivity analysis of the tuning protocol.
- ATLAS severity and sign sweeps with validation gates.

These limits define the validation path: broader event coverage, hardware timing, fixed-policy replication, and field replay.

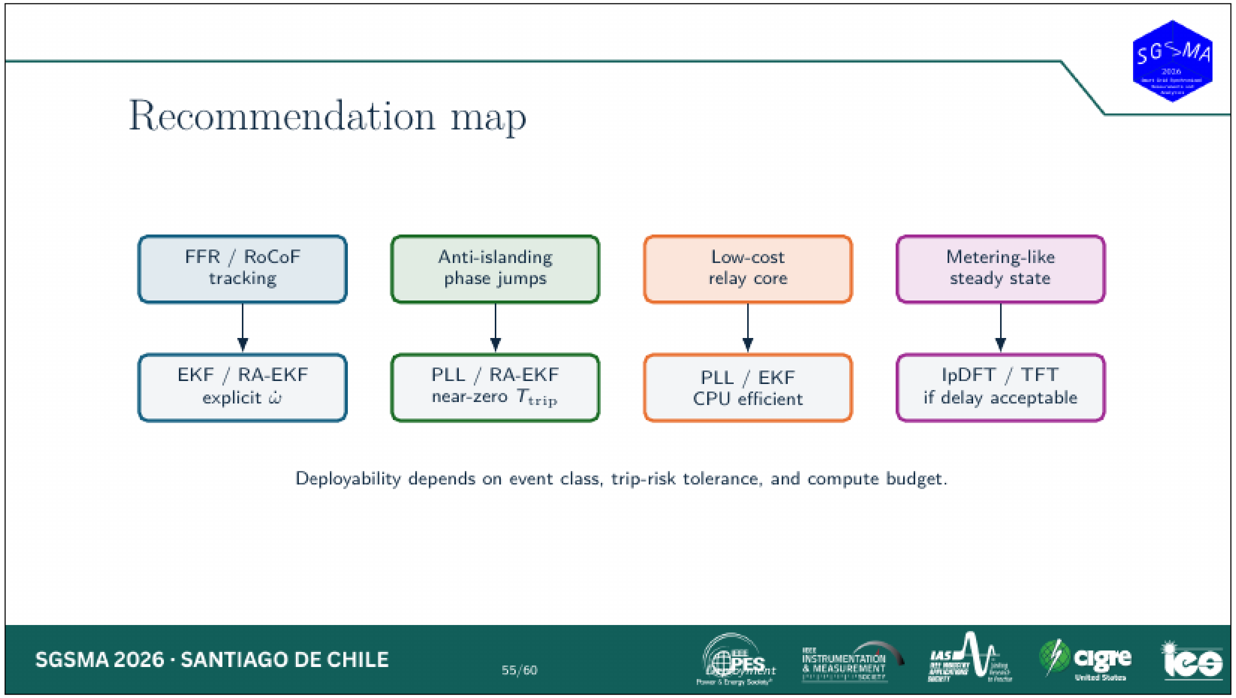
Validity limits and next steps

Say:

- On validity: the core means are five seeds, and the ATLAS sweeps are paper-grade at thirty.
- The next step is journal-grade at a hundred, plus the subspace and machine-learning methods.

→ *From here it becomes a recommendation.*

Key line: We are explicit about what is diagnostic and what is paper-grade.



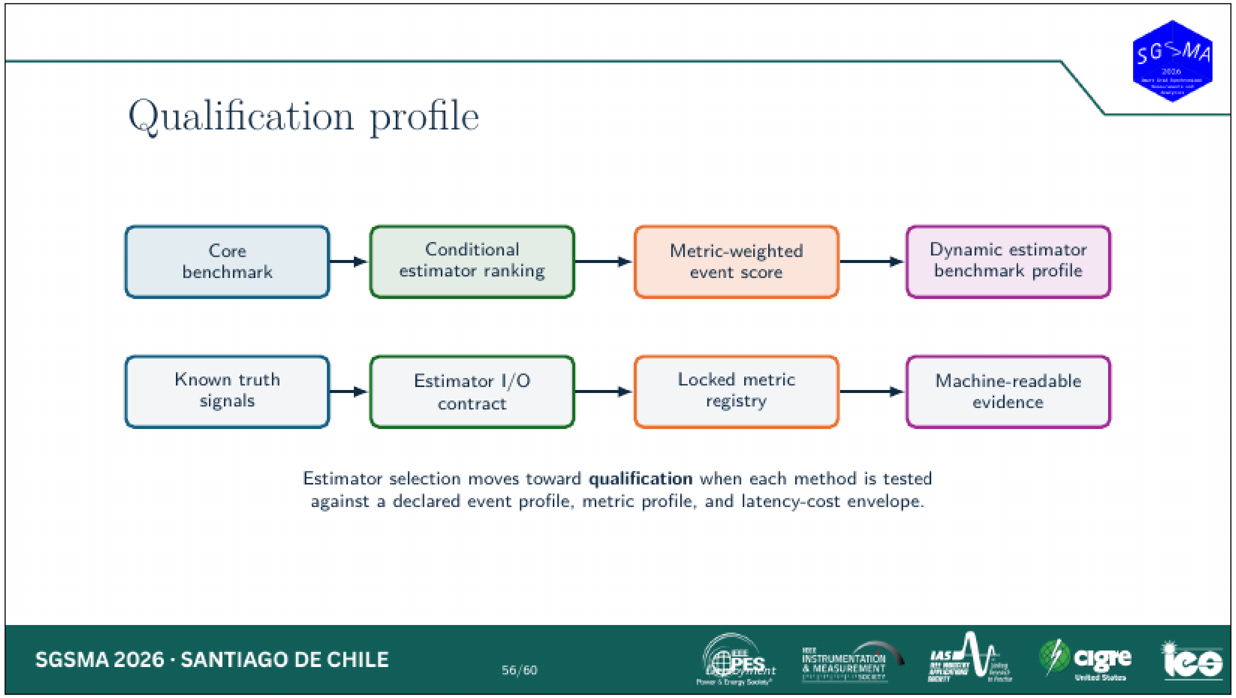
Recommendation map

Say:

- For fast frequency response and anti-islanding, RA-EKF is the strongest benchmark-backed choice.
- PLL and EKF are solid low-cost baselines, and IpDFT or TFT work where some observation delay is acceptable.

→ *And we wrap that into a qualification idea.*

Key line: We deliver a recommendation map, not a single ranking.



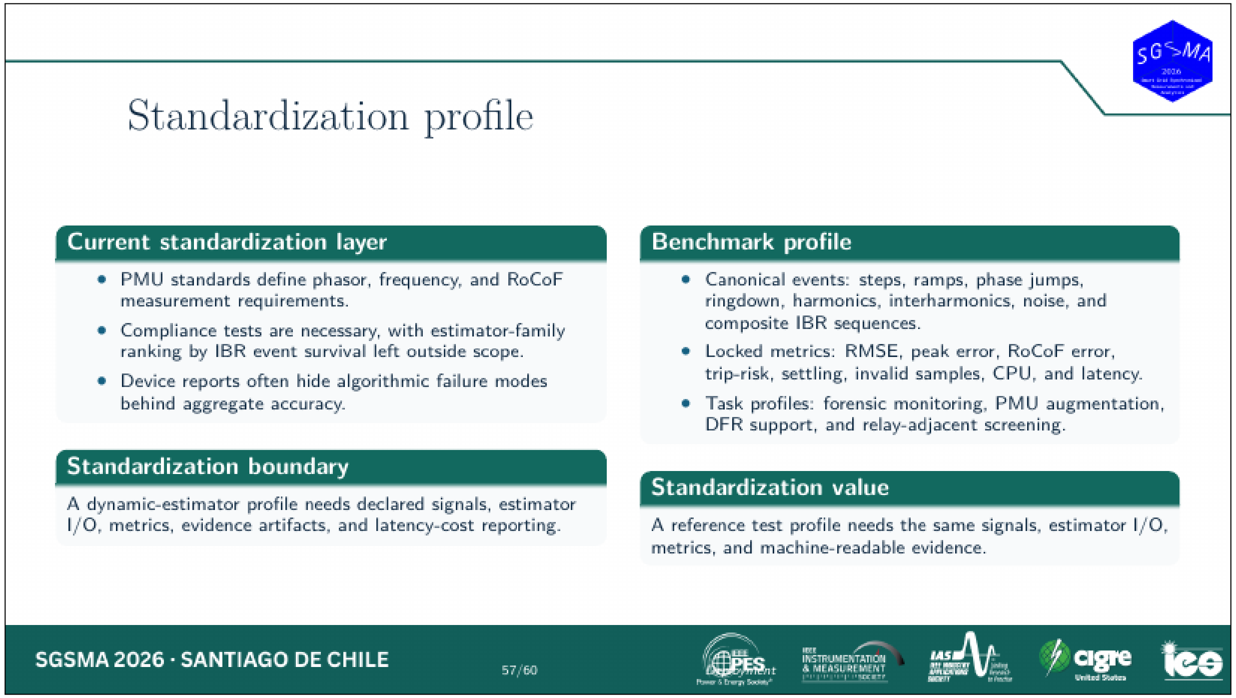
Qualification profile

Say:

- A real qualification profile should test transient recovery, trip exposure, compute feasibility, and standard-case RMSE.

→ *Reported in a standard, comparable form.*

Key line: Qualify on all dimensions together, not one at a time.



Standardization profile

Say:

- We propose reporting these dimensions explicitly, so any two estimators can be compared on equal terms.

→ *And when you truly need one number, here it is.*

Key line: Make the comparison reproducible and fair.

IBR Event Qualification Score

COMPOSITE EVENT SCORE: weighted event-class qualification metric

Score for estimator e

$$\text{IBR-ERS}(e) = 1 - \sum_{s \in \mathcal{S}} w_s \, \Phi_s(e),$$

$$\Phi_s(e) = \alpha \tilde{E}_s + \beta \tilde{P}_s + \gamma \tilde{T}_s + \delta \tilde{S}_s + \eta \tilde{C}_s + \lambda \tilde{L}_s.$$

Score composition

RMSE | peak | trip | settle | cost

$\tilde{E}_s, \tilde{P}_s$ dynamic and peak error
$\tilde{T}_s, \tilde{S}_s$ trip exposure and recovery
$\tilde{C}_s, \tilde{L}_s$ compute and latency penalties

Why a score is needed
RMSE, trip-risk, transient recovery, and cost lead to different decisions. A composite score makes the deployment objective explicit.

Protection profile
Raise γ, δ, λ when trip exposure, recovery, and latency dominate the risk model.

Monitoring profile
Raise α, β, η when reconstruction accuracy and compute budget dominate.

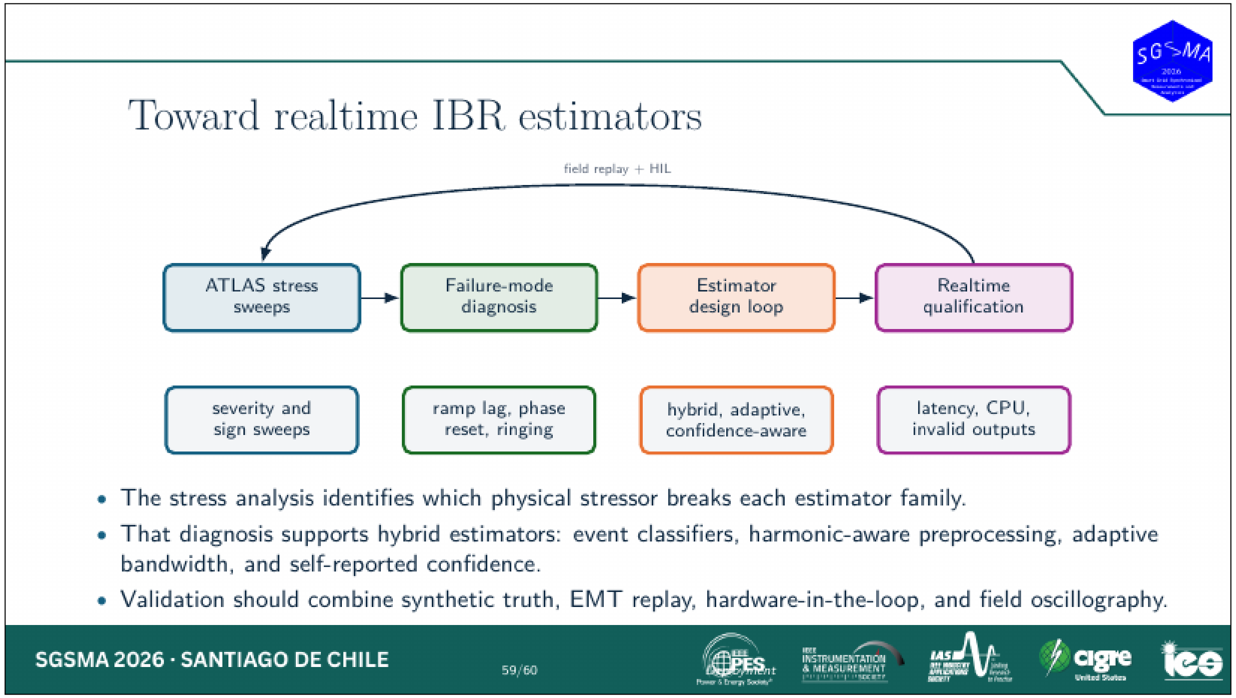
IBR Event Qualification Score

Say:

- When a single number is unavoidable, we define a composite, weighted, event-class score.

→ *Finally, where this work is going.*

Key line: One number when you need it, without hiding the dimensions.



Toward realtime IBR estimators

Say:

- Next: real-time estimators, integrating Simulink microgrid fault records, adding modern and machine-learning methods, C++ acceleration, and co-simulation with ANDES.

→ *Let me close.*

Key line: This is where the benchmark goes next.

Validation agenda and conclusion

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Validation agenda

- Full ATLAS sweeps: fixed policy, 100-run Monte Carlo, statistical rank-stability analysis.
- Composite qualification profiles for forensic, PMU-augmentation, and relay-adjacent tasks.
- Three-phase, unbalance, and converter-control interaction coverage.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize the full objective vector: frequency error, RoCoF, trip exposure, settling, CPU, latency.

Result: estimator choice is tied to event class, metric target, and compute envelope; estimator qualification must declare the stress regime, latency limit, and evidence depth.

Validation agenda and conclusion

Say:

- To conclude: there is no universal frequency estimator.
- The defensible choice depends on the event, the metric, and the compute budget.
- And OpenFreqBench gives the community a reproducible way to make that choice.

Key line: Thank you very much. I would be glad to take your questions.